**Relevant Positions**

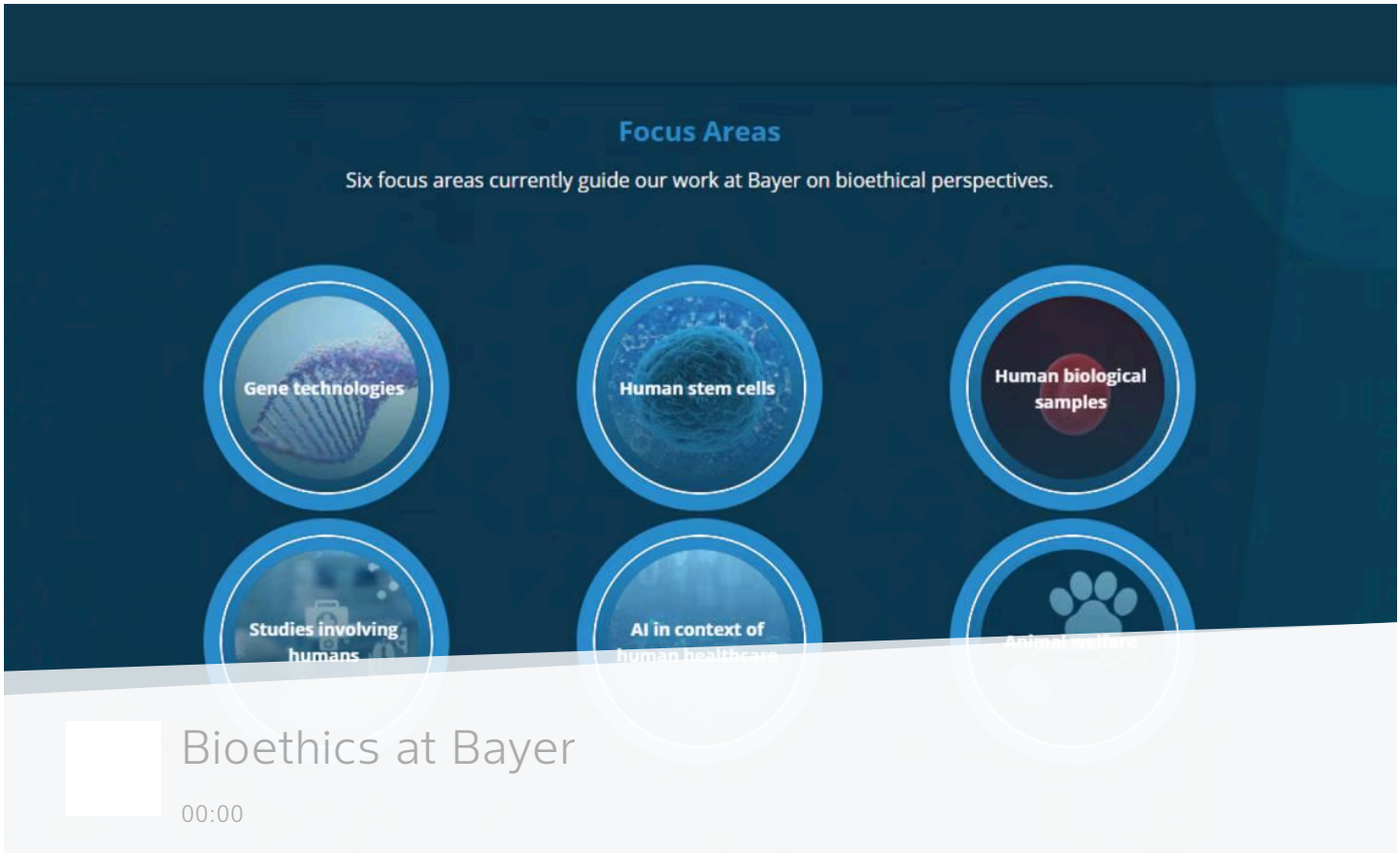
# Our Bioethical Principles

**Our six focus areas from a bioethical perspective**

- Gene Technologies
- Human Stem Cells
- Human Biological Samples
- Studies Involving Humans
- Artificial Intelligence (AI) in the Context of Human Healthcare
- Animal Welfare

As a life science company, committed to innovation in the areas of human health and agriculture, we align our actions with our mission ‘Health for all, Hunger for none’. Science and technology are advancing rapidly and creating new opportunities. We strive to utilize their potential for improving people´s lives. We are fully aware of our responsibility to deliver solutions and products to our patients and growers that are socially acceptable and sustainable.

- We are dedicated to conducting comprehensive risk/benefit assessment for any research and development work we undertake either within Bayer or with third parties
- We strictly comply with the applicable legal and regulatory frameworks as well as international convention
- We are committed to using innovative scientific approaches in an ethically responsible and transparent manner across all our divisions and to being thoughtful about the societal perception of science and use of technology




Thus, we have **defined Bioethical Principles** that set clear and company-wide* guardrails for R&D activities, innovation, and technology use. Our Bioethical Principles are grounded in our business ethics fundamentals and our **company values**. We continually review our high ethical standards and seek advice from a diverse group of leading experts, including our independent **Bioethics Council** members.

*The Bayer Bioethical Principles are valid for Bayer AG and all Bayer group companies.

# Gene Technologies

### Use of Gene Technologies for Healthcare:

At Bayer, we consider gene technologies to be powerful tools in drug discovery and development. We use gene technology tools to study human diseases, improve manufacturing processes, and develop new treatments, including regenerative gene therapies. Bayer believes that gene therapies have a promising potential to bring tremendous value to patients suffering from rare monogenic diseases (such as Pompe disease) or to broader patient populations with complex pathway diseases (like heart failure). As such gene technologies provide novel treatment options for conditions for which there is a lack of alternative treatments, or where the current standard of care only addresses symptoms to different degrees. Bayer also recognizes the potential of gene technologies to develop more targeted, personalized treatment options.

Guiding Principles for use of Gene Technologies for Healthcare

Definition of Gene Technologies for Healthcare:

## Discover More about Gene Technologies for Healthcare:

Gene Editing >

Gene Therapy >

AAV Therapeutics >

## Use of Gene Technologies for Agriculture:

In our Crop Science business, we use gene technology tools to produce seed, plants, and microbes with beneficial traits that provide nutritional, agronomic, and economic benefits that significantly reduce the environmental footprint and impact of agriculture while increasing productivity.

In the face of challenges posed by climate change, gene technologies provide rapid methods of developing and selecting for more resilient crops, which complement conventional breeding methods.

We are conscious of potential environmental and ethical considerations related to agricultural applications of gene technologies. We take the principles of stewardship seriously, for example by establishing guidelines for our own sites and those of our seed producers, which must be followed. These measures include protecting locally important landraces. Bayer is committed to providing continuous, collaborative support to farmers. This includes the co-design of sustainable farming practices, stewardship, quality management activities, and training programs related to agricultural applications of our gene technologies.

Guiding Principles for use of Gene Technologies for Agriculture

Definition of Gene Technologies for Agriculture

## Discover More about Gene Technologies for Agriculture:

Genome Editing in Plant Breeding >

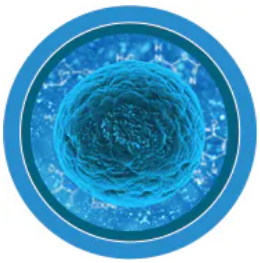
GMO and Biotechnology >

Product Stewardship >

Genetically Modified Crops I Bayer´s contribution to a fact-based public discourse:

> Download Executive Summary (419 KB)

# Human Stem Cells



Human stem cells have great potential for regenerative medicine and drug discovery. There are two types of pluripotent stem cells which can turn into virtually any kind of cell in the human body: embryonic stem cells and induced pluripotent stem cells which are generated from adult cells e.g., blood cells. Bayer wants to use these cells to develop regenerative therapies for severe and life-threatening diseases, such as Parkinson's or heart failure. In addition, Bayer also utilizes those cells to test new drugs on human cells. However, we are aware that some people have concerns about using human embryonic stem cells.

Guiding Principles for use of Human Stem Cells

Definition of Induced Pluripotent Stem Cells

## Discover More about we use Human Stem Cells at Bayer

Stem Cells >

# Human Biological Samples

Research using human biological samples (HBS), such as tissue and blood, is performed to enhance the knowledge of human diseases, improve diagnosis and to discover and develop new treatments.
Conducting research on human biological samples is critical for the development of safe and efficacious new drugs and therapies.

Guiding Principles for use of Human Biological Samples

# Studies Involving Humans

Studies in humans play a vital role in the Pharmaceutical and Consumer Health product development process as they provide essential insights into the safety and effectiveness of new products.
At Bayer, we pay utmost attention to the individual's well-being, dignity, safety, and rights. Our ethical considerations for studies in humans revolve around four fundamental principles:

- Obligation to consider the benefit for the study participant or to help others (Beneficence)
- Obligation to avoid causing harm to the participant or to others (Nonmaleficence)
- Concerns that participants are treated with fairness and equity (Justice)
- Respect for Autonomy of the participant.

Guiding Principles for Conducting Studies Involving Humans

## Discover More about Studies Involving Humans:

Ethics in Clinical Trials >

Clinical Trials >

Clinical trial transparency policy >

# Artificial Intelligence (AI) in the Context of Human Healthcare

AI is becoming increasingly important for bringing new advances in areas such as the life sciences and many areas of society. At Bayer, we believe that AI is an important technical enabler for many aspects of our business and operations. For example, AI already plays a crucial role in drug discovery and development, personalized medical treatments and improving diagnostics. Since AI is constantly evolving, we will keep refining our bioethical stance and principles to ensure responsible and thoughtful use of this technology.

Guiding Principles for the use of Artificial Intelligence (AI) in the Context of Healthcare

personal autonomy, strive to promote individual and collective wellbeing and preventing harm, while also striving for inclusion in our data sets, fairness and accountability.

- We always comply with all applicable laws, regulations, and professional codes of conduct.
- We regularly cooperate with researchers, regulatory bodies, and other relevant professional organizations to jointly develop trustworthy AI-enabled solutions.
- We hold companies that Bayer partners with or whose products or services are procured to the same standard.
- We strive for clear relevance, provenance, diversity, and sufficient quality in data sources.
- We verify the data sets used for AI training are representative of the environment the AI system will be deployed in.
- We do not use poor, incomplete or non-verified data that may produce incorrect or biased results in our AI applications.
- We will protect human autonomy in AI solutions that we develop, sell or use. We will develop, dedicate, and put into practice internal resources, structures, and safeguards to support proper internal oversight, inspections, and maintenance of AI solutions.

## Definition for Artificial Intelligence (AI) in the Context of Human Healthcare

Artificial intelligence (AI) AI is the development of computer systems that can perform tasks requiring human intelligence, like analyzing data, recognizing patterns, and making decisions. It involves creating algorithms and models to enable machines to learn and mimic human cognitive abilities.

## Discover More about Artificial Intelligence in the Context of Human Healthcare:

AI in Pharma >

# Animal Welfare

Animal studies still play an important role in ensuring that the benefits of our products outweigh the potential risks to humans, animals and the environment, and therefore remain part of the current standard data requirements and approval process of our products by regulatory authorities.

Guiding Principles for Animal Welfare in Research and Development

## Discover More about Animal Welfare:

The Essential Role of Animal Studies >

The Current State of Animal Studies >

Promoting Alternative Methods >

Links

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